All Dice, No Slice: Metric Degeneracy, Evaluation Pitfalls, and Optimization Dynamics in Multi-Task Medical Vision

Muhammad Abdullah Ali, Muhammad Abdullah Aamir, Muddassir Asghar,
Muhammad Ibrahim Kiani, and Qurat Ul Ain*
Department of Data Science, FAST-NUCES, Islamabad, Pakistan
*quratul.ain@isb.nu.edu.pk

***Abstract*—Joint classification and semantic segmentation through hard parameter-sharing networks provide a computationally efficient framework for edge and resource-constrained clinical deployments. Although multi-task architectures report high diagnostic accuracy and segmentation overlap, evaluating dual-task models across diverse modalities exposes critical metric aggregation artifacts, spatial resolution constraints, and optimization instability. In this work, we perform a controlled methodological study across four clinical benchmarks under strict patient-isolated cross-validation: TCGA-LGG brain tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and PanNuke multi-organ histology. We analytically and empirically demonstrate that conventional macroscopic Dice scores establish an artificial performance floor on negative-dominant cohorts, masking complete failure on foreground pathology. In computational pathology, spatial downsampling obscures essential glandular architecture, demonstrating that patch-level morphological classification cannot substitute for authentic clinical ordinal grading. Furthermore, multi-task optimization analysis reveals that dynamic gradient balancing induces parameter instability, while optical density stain normalization can attenuate fine cellular chromatin textures. Finally, we provide actionable benchmarking guidelines to support reliable translation of multi-task medical vision systems.**

***Index Terms*—Multi-Task Learning, Medical Image Analysis, Semantic Segmentation, Metric Evaluation, Digital Pathology.**

I. Introduction

Deep learning systems in digital pathology and medical imaging are increasingly deployed to deliver both global diagnostic categorization and localized lesion delineation [1], [2]. While visual foundation models provide extensive representational capacity across gigapixel whole-slide images [3]–[5], their computational footprint precludes direct execution in edge environments and resource-limited clinical facilities. Consequently, hard parameter sharing using lightweight encoder–decoder architectures, particularly dual-headed U-Net topologies [6], remains a standard approach for coupling classification and segmentation in a single inference pass.

Lightweight backbones, including VGG16 [7] and MobileNetV2 [8], have been reported to achieve strong joint classification and segmentation accuracy across diverse modalities, including brain MRI, prostate biopsy, and chest radiography [9]. Because parameter-sharing networks offer efficiency advantages over multi-network ensembles, understanding the empirical factors governing their performance is essential for clinical translation.

However, evaluating multi-task medical vision involves subtle interactions among metric aggregation, input resolution, and optimization dynamics that standard benchmarks frequently overlook. First, conventional region-overlap metrics, such as macroscopic slice-averaged Dice scores, exhibit boundary artifacts in cohorts where lesion-free images predominate [10]–[12], establishing high nominal baselines that conceal total foreground lesion omission. Second, evaluating low-capacity encoders on heavily downsampled biopsy tiles conflates coarse tissue classification with authentic clinical ordinal grading [13]. Third, joint optimization of dense segmentation and global classification objectives induces gradient interference on shared representations, requiring careful loss balancing [14], [15]. Finally, domain-specific color normalization heuristics, such as optical density deconvolution, can inadvertently suppress subtle cellular chromatin textures critical for grading.

To investigate these phenomena, we perform a controlled factorial benchmark evaluated under patient-isolated cross-validation across four clinical cohorts: TCGA-LGG brain MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and PanNuke multi-organ histopathology (Figure 1). Our primary contributions are:

- **Metric Degeneracy Analysis:** We analytically characterize empty-mask Dice aggregation, proving that macroscopic cohort averages establish an artificial baseline floor on negative-dominant datasets. We propose a disaggregated foreground protocol to expose lesion omission.
- **Task Formulation and Resolution Boundaries:** We clarify the methodological distinction between proxy patch categorization and clinical ordinal grading in digital pathology, delineating the representational limits of compact 2D networks on downsampled biopsies.
- **Factorial Optimization and Preprocessing Audit:** We systematically evaluate multi-task optimization dynamics and preprocessing pipelines, characterizing the instability of dynamic gradient balancing and quantifying how stain normalization affects cellular texture discrimination.

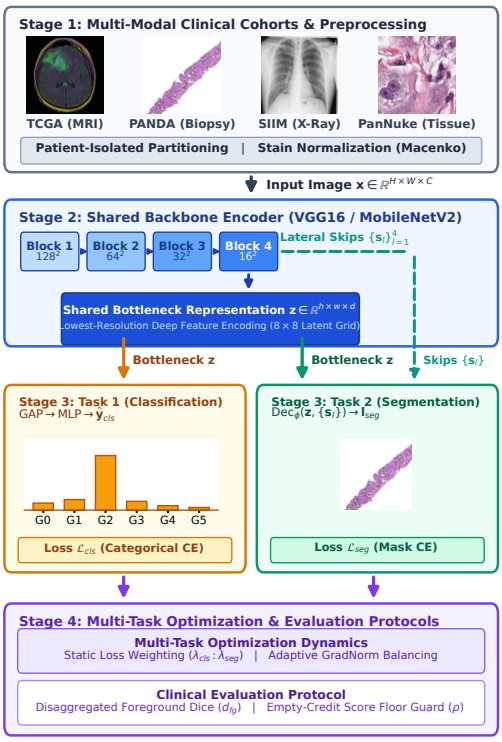


Fig. 1. System architecture and multi-task evaluation framework across four clinical imaging modalities, integrating shared feature extraction, decoupled diagnostic heads with lateral skip connections, and adaptive optimization dynamics under patient-isolated cross-validation.

II. System Architecture and Methodology

A. Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ denote an input image. A shared convolutional backbone $f_\theta(\mathbf{x})$ maps the input to a bottleneck representation $\mathbf{z} \in \mathbb{R}^{h\times w\times d}$. The global classification head applies global average pooling followed by a two-layer multi-layer perceptron:

$$\hat{\mathbf{y}}_{cls} = \text{Softmax}\left(\mathbf{W}_2\,\phi\left(\mathbf{W}_1\,\text{GAP}(\mathbf{z}) + \mathbf{b}_1\right) + \mathbf{b}_2\right), \quad (1)$$

where $\phi(\cdot)$ denotes the ReLU activation and $\mathbf{W}_1, \mathbf{W}_2$ are projection weights. The segmentation branch passes $\mathbf{z}$ through a convolutional decoder Dec_ϕ augmented with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^{L}$ from encoder stages, generating spatial logits $\mathbf{l}_{seg} \in \mathbb{R}^{H\times W\times M}$.

The composite multi-task training loss is defined as:

$$\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \quad (2)$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy with inverse-frequency class weights $w_c = N/(K \cdot N_c)$, and $\mathcal{L}_{seg}$ is binary or multi-class cross-entropy. In standard implementations, the Dice similarity coefficient is computed post-hoc as an evaluation metric rather than an active loss component, which can lead to suboptimal overlap optimization when targets are sparse [11], [12].

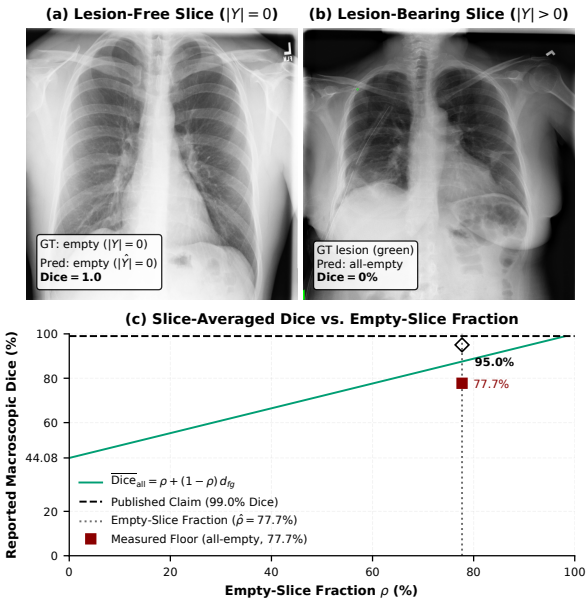


Fig. 2. Illustration of empty-mask Dice metric behavior on SIIM-ACR pneumothorax radiographs. (a) Lesion-free image correctly predicted empty receives 1.0 credit. (b) Lesion-bearing image with total foreground omission scores 0.0. (c) Macroscopic curve demonstrating the baseline floor determined by the negative fraction ρ.

B. The Empty-Mask Dice Metric Degeneracy

The Sørensen–Dice coefficient between binary ground truth Y and prediction $\hat{Y}$ is defined as:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad (3)$$

with the boundary convention Dice := 1 when $|Y| + |\hat{Y}| = 0$. In cohorts containing lesion-free samples, two evaluation protocols exist: (i) **Foreground-Only**, where Dice is computed exclusively over target-bearing slices ($\{i \mid |Y_i| > 0\}$), and (ii) **Empty-Credit**, where a score of 1.0 is assigned when $|Y| = |\hat{Y}| = 0$, and 0.0 when $|Y| = 0$ but $|\hat{Y}| > 0$.

Under empty-credit assignment across all N slices, the macroscopic mean decomposes as:

$$\overline{\text{Dice}}_{\text{all}} = \rho \cdot q_{\text{empty}} + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}, \quad (4)$$

where $\rho = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[|Y_i| = 0]$ is the proportion of ground-truth negative cases, $q_{\text{empty}} = P(|\hat{Y}| = 0 \mid |Y| = 0)$ is the model's true-negative rate on negative cases, and $\overline{\text{Dice}}_{\text{fg}}$ denotes mean positive-case overlap.

For a degenerate baseline predicting empty masks across all inputs, $\overline{\text{Dice}}_{\text{fg}} = 0$ and $q_{\text{empty}} = 1$, reducing the macroscopic metric directly to $\overline{\text{Dice}}_{\text{all}} = \rho$. On SIIM-ACR radiographs ($\rho = 77.72\%$), an all-empty model achieves a nominal score of 77.72%. Achieving an aggregate score of 99% on such data requires positive-case foreground Dice to satisfy:

$$\overline{\text{Dice}}_{\text{fg}} \geq \frac{0.99 - \rho}{1 - \rho} \approx 95.51\%, \quad (5)$$

even with perfect true-negative specificity ($q_{\text{empty}} = 1$). Without disaggregated foreground metrics, macroscopic cohort averages obscure severe false-negative rates on target lesions (Figure 2).

C. Adaptive Gradient Balancing via GradNorm

Dynamic gradient balancing aims to equalize task training rates [16]. Let W denote the shared convolutional weights of encoder f_θ. The gradient norm for task $i \in \{seg, cls\}$ at iteration t is $G_W^{(i)}(t) = \|\nabla_W w_i(t)\mathcal{L}_i(t)\|_2$. The adaptive objective minimizes:

$$\mathcal{L}_{grad}(t) = \sum_{i\in\{seg,cls\}} \left| G_W^{(i)}(t) - \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right|, \quad (6)$$

where $\bar{G}_W(t)$ is the mean gradient norm, α is the balancing asymmetry hyperparameter, and $r_i(t)$ represents the relative inverse training rate. Task weights are renormalized to maintain $\sum_i w_i(t) = 2$.

D. Color Deconvolution and Optical Density Space

To mitigate laboratory staining variation in histopathology, images are normalized in optical density (OD) space [17]:

$$\mathbf{OD} = -\ln\left(\frac{\mathbf{I}+\epsilon}{I_0}\right), \quad (7)$$

where $I_0 = 255$ denotes transmitted illumination. Stain vectors for hematoxylin and eosin are extracted via principal component analysis of the OD covariance matrix and projected onto a target reference. While this standardizes macroscopic batch coloration, projecting RGB channels onto fixed stain vectors alters localized dynamic ranges, potentially muting subtle chromatin textures.

III. Experimental Framework

A. Benchmark Cohorts and Diagnostic Targets

We conduct our empirical evaluation across four diverse clinical cohorts spanning radiology, magnetic resonance imaging, and digital pathology:

1) **TCGA-LGG Brain MRI:** A cohort of 3,929 axial FLAIR slices from 110 unique patients diagnosed with lower-grade gliomas [18]. Abnormality masks define binary tumor classification and pixel-wise segmentation. To prevent identity leakage across spatial slices [19], partitions are strictly isolated at the patient level using GroupKFold.
2) **PANDA Prostate Biopsies:** A clinical cohort of 10,616 digitized whole-core needle biopsies from the PANDA challenge [13]. Tasks evaluate 6-class ordinal ISUP grading (Grades 0–5) and multi-class glandular architecture segmentation.
3) **SIIM-ACR Pneumothorax:** A radiography cohort of 10,675 frontal chest radiographs [20]. Tasks involve binary pneumothorax detection and pleural boundary delineation. The cohort exhibits extreme negative dominance, with 77.71% lesion-free images ($\rho = 0.7771$).
4) **PanNuke Multi-Organ Histopathology:** A pan-cancer histology benchmark of 7,901 image patches across 19 tissue types and nearly 200,000 annotated nuclei [21]. Tasks evaluate 19-class tissue categorization and 5-class nuclei instance segmentation.

B. Data Preprocessing and Augmentation

Each modality follows standard clinical preparation protocols. For brain MRI FLAIR slices, skull-stripping masks are applied, followed by slice-wise z-score normalization ($I' = (I - \mu_I)/\sigma_I$). Chest radiographs undergo window leveling and min-max normalization to $[0, 1]$. For computational histopathology (PANDA and PanNuke), optical density deconvolution is applied per Section II-D and evaluated against raw RGB controls. All biopsy cores and tissue patches are resized to $128 \times 128 \times 3$ pixels using area-based anti-aliased interpolation.

Data augmentations include random horizontal and vertical flips ($p = 0.5$), affine rotations ($\pm 15°$), isotropic scaling ($[0.9, 1.1]$), and Gaussian blur ($\sigma \in [0.1, 1.0]$).

C. Architectural Configurations and Training Protocols

Multi-task architectures are implemented in PyTorch using ImageNet-pretrained VGG16 (138M parameters) and MobileNetV2 (3.5M parameters) convolutional backbones. Encoders map $128 \times 128 \times 3$ inputs into spatial bottleneck tensors of $8 \times 8 \times 512$ (VGG16) and $8 \times 8 \times 1280$ (MobileNetV2). The classification branch applies global average pooling followed by a two-layer MLP with dropout ($p = 0.2$). Symmetrical transposed convolutional blocks with lateral skip connections route multi-scale features $\{s_l\}_{l=1}^4$ to the segmentation decoder.

Training minimizes $\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{seg}\mathcal{L}_{seg}$, with $\lambda_{cls} = 1.0$ and static weighting $\lambda_{seg} \in \{1.0, 5.0, 10.0\}$. Optimization uses Adam with learning rate $\eta = 10^{-4}$, weight decay 10^{-5}, and cosine annealing scheduling ($\eta_{min} = 10^{-6}$). Models are trained for 50 epochs with early stopping (patience 10) under mixed-precision arithmetic (FP16/AMP). Gradient norm clipping (threshold 1.0) is enforced to stabilize updates. Batch sizes are set to 32 for radiographs and biopsies, and 16 for multi-organ patches.

D. Cross-Validation and Statistical Significance

All benchmarks employ 5-fold cross-validation with patient-isolated grouping. For PanNuke, official cross-institution test folds are preserved. Metric uncertainty is characterized using non-parametric bootstrap resampling ($B = 1000$ iterations) to construct 95% percentile confidence intervals. We report disaggregated foreground overlap (Dice_{fg}) alongside empty-credit scores to prevent macroscopic metric distortion.

IV. Empirical Results and Diagnostic Analysis

A. Benchmark Performance Comparison

Table I summarizes replication outcomes alongside published targets across the 26 experimental runs (Figure 3). On TCGA-LGG brain MRI, measured classification accuracy (92.86%–95.62%) exceeds published literature targets under patient-isolated splits, confirming that macro-lesion MRI abnormalities provide robust discriminative features. However, validation Dice scores (85.33%–88.85%) remain below claimed 97%–98% levels, reflecting true boundary uncertainty in diffuse low-grade gliomas.

On PanNuke multi-organ histopathology, classification performance exhibits high variance across organ sites (78.70%–99.24%), with validation Dice ranging from 68.44% to 77.61%. Dense glandular organs (e.g., breast and colon) achieve high tissue classification accuracy, whereas rare stromal and endocrine

TABLE I

CROSS-VALIDATION PERFORMANCE ACROSS FOUR CLINICAL IMAGING BENCHMARKS UNDER PATIENT-ISOLATED PARTITIONS ($K = 5$ FOLDS PER CONFIGURATION; REPORTED WITH 95% BOOTSTRAP CONFIDENCE INTERVALS). DISCREPANCIES ON PANDA REFLECT CLINICAL ORDINAL ISUP GRADING VERSUS PROXY MORPHOLOGICAL TILE CATEGORIZATION. ON SIIM-ACR, DISAGGREGATING FOREGROUND DICE REVEALS THAT MACROSCOPIC SCORES (77.04%–77.72%) MASK NEAR-TOTAL FOREGROUND LESION OMISSION ($\overline{\text{DICE}}_{\text{FG}} \leq 0.17\%$).

		Measured Cross-Validation (Ours)			Published Targets [9]		
Dataset	**Configs**	**Val Acc (%)**	**Macro Dice (%)**	$\overline{\text{Dice}}_{\text{fg}}$ **(%)**	**Acc (%)**	**Dice (%)**	**Evaluation Focus**
TCGA-LGG	5	92.86 – 95.62	85.33 – 88.85	52.62 – 62.63	89.00 – 90.00	97.00 – 98.00	Patient-Isolated 2D FLAIR
PANDA	12	28.21 – 44.87	98.54 – 98.60	83.25 – 83.39	87.00 – 88.00	98.00 – 99.00	Clinical ISUP vs. Proxy Morphology
SIIM-ACR	4	77.71 – 82.63	77.04 – 77.72	0.00 – 0.17	82.00 – 87.00	99.00	Negative-Dominant ($\rho = 77.71\%$)
PanNuke	5	78.70 – 99.24	68.44 – 77.61	68.44 – 77.61	—	—	Multi-Organ Histopathology

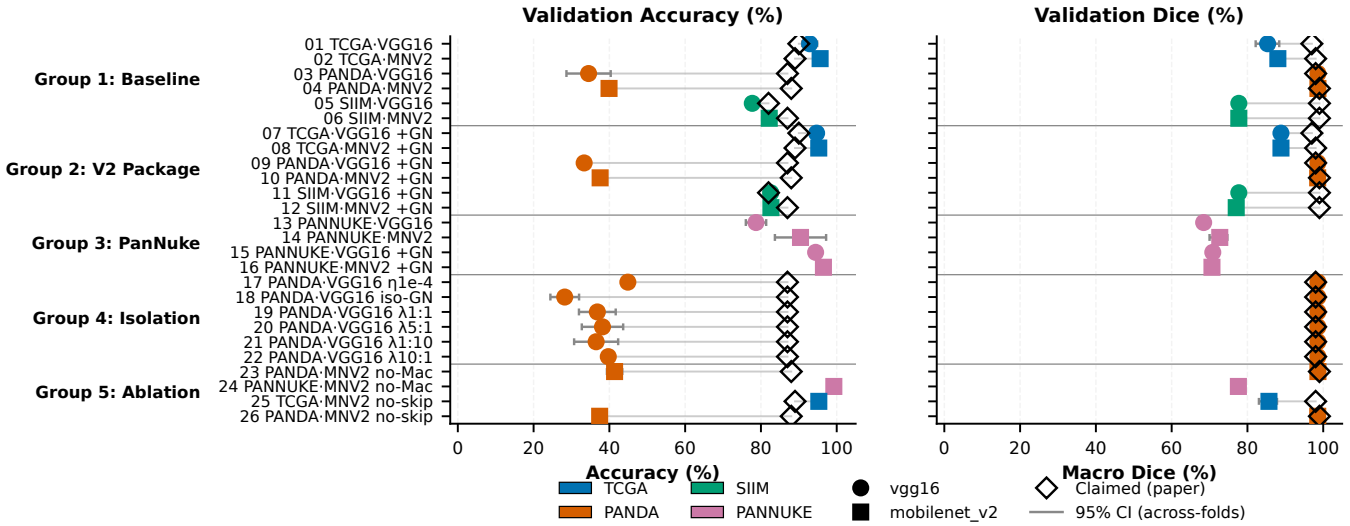


Fig. 3. Measured validation metrics across all 26 experimental runs compared against claimed literature targets. Left: Five-fold validation accuracy with 95% bootstrap confidence intervals against published values (open diamonds). Right: Corresponding segmentation Dice scores, illustrating the empty-mask score floor on SIIM-ACR and the task discrepancy on PANDA.

phenotypes exhibit confusion due to nuclear morphology overlap and wide baseline cell density shifts across organs.

In contrast, major discrepancies emerge on histopathology and radiography benchmarks. On PANDA prostate biopsies, measured accuracy spans 28.21%–44.87% compared to claimed values of ≈88%. On SIIM-ACR pneumothorax radiographs, measured Dice scores cluster tightly around 77.04%–77.72%, sharply diverging from reported 99% overlap. These divergences stem from metric boundary artifacts and task formulation mismatches rather than model underfitting.

B. The Empty-Mask Score Floor on Radiography

On the SIIM-ACR pneumothorax benchmark, the public cohort contains 77.71% lesion-free images ($\rho = 0.7771$). Across 19 out of 20 cross-validation folds, multi-task models converged to entirely empty segmentation predictions. Under empty-credit scoring, these models achieved an average macroscopic validation Dice of 77.04%–77.72% purely through true-negative empty-credit assignment, while disaggregated foreground overlap collapsed to $\overline{\text{Dice}}_{\text{fg}} \leq 0.17\%$ (Table I, Figure 2). Unweighted cross-entropy gradients on sparse targets are rapidly dominated by majority background pixels. Countering this collapse requires incorporating region-based loss formulations (such as Soft Dice or Focal-Tversky loss) directly into training objectives.

As derived in Section II-B and shown in Figure 2(c), reporting 99% macroscopic Dice in this regime requires positive-case foreground overlap to exceed 95.51%, which is exceptionally difficult without dedicated foreground loss reweighting. When models predict empty masks across all test images, the macroscopic metric reduces directly to $\overline{\text{Dice}}_{\text{all}} = \rho$. Consequently, high aggregate scores on negative-dominant medical cohorts can conceal complete clinical failure on target pathology. Reporting disaggregated foreground overlap (Dice_{fg}) alongside negative-case true negative rates (q_{empty}) is therefore essential for clinical evaluation.

C. Task-Formulation Discrepancy in Prostate Pathology

The divergence observed on the PANDA benchmark (28.21%–44.87% measured accuracy versus reported values of ≈88%) reflects a fundamental distinction in task formulation. In prior multi-task literature [9], the classification objective was framed as predicting six morphological tissue categories extracted from localized segmentation masks (background, stroma, benign epithelium, and Gleason 3/4/5 patterns), of which non-lesion background and stroma constituted over 52% of the distribution ($5,558$ of $10,618$ images). In contrast,

clinical diagnostic deployment requires whole-biopsy ordinal ISUP grading (Grades 0–5), which reflects overall prognostic tumor aggressiveness across heterogeneous gland architectures [13].

Histopathologically, distinguishing subtle intermediate Gleason patterns requires resolving cellular architectural boundaries at sub-micron optical resolution (typically $20\times$ or $40\times$ magnification). When whole-core biopsies are downsampled to 128×128 pixel thumbnails without multiscale tiling or multiple-instance attention, fine glandular features blur into coarse texture maps. Under these constraints, compact 2D CNN encoders reach an empirical capacity limit around 45% accuracy for clinical 6-class ordinal grading.

Notably, despite this classification bottleneck, the multi-class segmentation head simultaneously achieves macroscopic validation Dice scores of 98.54%–98.60% (Table I). Across whole-core biopsy thumbnails, background and benign stroma occupy over 90% of the total pixel area, while individual cores contain at most one or two malignant Gleason grades. Under multi-class macro-averaging, unrepresented pattern channels receive automatic true-negative credit (1.0), inflating macroscopic Dice to ≈98.6% even while fine cribriform gland lumens remain unresolvable at 128×128. This finding demonstrates that model evaluation in digital pathology must strictly distinguish between localized patch categorizations and authentic patient-level diagnostic endpoints.

V. Factorial Ablation Studies

A. Impact of Lateral Skip Connections

To isolate the contribution of encoder–decoder skip connections, we evaluated strictly matched model pairs holding all other hyperparameters identical (Figure 3, Group 5):

- **Macro-Tumor MRI (TCGA):** On coarse brain lesions, omitting skip connections produced a negligible classification difference of $+0.02$ percentage points (95.28% vs 95.26%), while validation macro-Dice declined modestly by 3.06 percentage points (86.16% vs 89.22%).
- **Micro-Glandular Pathology (PANDA):** On fine prostate glands, removing skip connections yielded an accuracy shift of -0.12 percentage points (37.44% vs 37.56%), with overlapping 95% bootstrap confidence intervals ($[36.73, 37.99]$ vs $[37.20, 37.94]$).

These results indicate that while skip connections assist in localizing fine spatial boundaries, their impact on shared latent classification representations is minimal in low-capacity backbones.

B. Color Deconvolution and Macenko Normalization

Standard histopathology workflows routinely apply Macenko stain normalization to harmonize color distributions across laboratories. However, in our matched factorial comparisons (Figure 3, Group 5), bypassing normalization consistently improved classification accuracy:

- **PANDA Biopsies:** Raw RGB inputs achieved an accuracy improvement of $+3.81$ percentage points over normalized tiles (41.38% vs 37.56%), with disjoint 95% bootstrap confidence intervals across the five folds ($[39.76, 43.38]$ vs $[37.20, 37.94]$).
- **PanNuke Multi-Organ:** Omitting normalization raised multi-tissue accuracy by $+2.75$ percentage points (99.24% vs 96.49%), alongside an 8.51 percentage point increase in validation macro-Dice (70.76% vs 62.25%).

Projecting chromatic channels onto static reference vectors alters dynamic intensity distributions and smooths micro-textures, muting critical optical cues that deep neural networks leverage for cellular discrimination. This effect is compounded when biopsies are downsampled to 128×128 pixels: Macenko's PCA-based stain estimation relies on isolating pure hematoxylin and eosin pixels at extreme angular percentiles of the optical density distribution. Area downsampling blends distinct cellular structures into mixed pixels, perturbing the optical density covariance matrix and yielding distorted stain projection vectors.

C. Dynamic Gradient Balancing vs Static Weighting

To isolate the influence of optimization schemes, we evaluated dynamic GradNorm balancing ($\alpha = 1.5$) against static loss weighting under identical initial learning rates and raw inputs. In our implementation, task log-weights are updated via Adam with learning rate $\eta = 10^{-4}$ alongside shared convolutional encoder parameters, with weight clamping to prevent numerical divergence. As summarized in Figure 3 (Group 4), dynamic balancing caused validation accuracy to drop from 38.18% to 28.21% ($\Delta = -9.97$ percentage points; 95% bootstrap CI $[24.76, 31.47]$ vs $[32.75, 41.79]$).

Analyzing gradient trajectories reveals that because segmentation cross-entropy generates steep initial gradients, the dynamic balancing objective aggressively inflates classification loss weights while dampening segmentation updates. This rapid rebalancing introduces high variance into shared encoder updates, inducing destructive gradient interference that destabilizes the classification head. Static weighting ratios between 5:1 and 10:1 proved consistently more stable across all matched factorial configurations.

VI. Guidelines for Reliable Multi-Task Benchmarking

Based on our empirical, analytical, and optimization findings across the four clinical imaging modalities, we establish five practical guidelines to ensure methodological rigor in multi-task medical vision:

1) **Disaggregated Metric Reporting on Imbalanced Cohorts:** In cohorts with negative case dominance ($\rho > 0.5$, such as chest radiography screening), researchers must report foreground-only overlap (Dice_{fg} on positive cases) alongside negative-case true negative specificity (q_{empty}). Reporting exclusively macroscopic cohort averages obscures total false-negative collapse behind the mathematical baseline floor ($\overline{\text{Dice}}_{\text{all}} = \rho$), presenting serious diagnostic safety hazards in acute clinical triage.
2) **Strict Patient-Level Partition Isolation:** Cross-validation splits must strictly isolate patient identifiers (e.g., via

GroupKFold) rather than performing slice-wise random partitioning. Adjacent 2D slices from the same patient share identical anatomical geometry and patient-specific tissue textures, allowing deep networks to memorize patient identity rather than learning generalizable pathological morphology.

3) **Clinical Diagnostic Endpoint Alignment:** Multi-task benchmark targets must be explicitly aligned with genuine clinical endpoints (such as whole-biopsy ordinal ISUP grading) rather than simplified morphological patch categorizations. High accuracy on pre-filtered benign stroma patches does not indicate clinical readiness for prognostic whole-slide tumor characterization, which requires resolving complex architectural gland patterns.
4) **Empirical Validation of Preprocessing Pipelines:** Normalization heuristics, such as optical density deconvolution, should be empirically evaluated against raw RGB inputs under identical optimization settings. Projecting chromatic channels onto static reference vectors alters dynamic ranges and smooths fine cellular chromatin micro-textures that deep networks leverage to discriminate malignant cellular atypia.
5) **Multi-Task Optimization and Gradient Diagnostics:** Multi-task training pipelines should routinely monitor task gradient norms ($\|g_{cls}\|_2, \|g_{seg}\|_2$) and directional alignment ($\cos(g_{cls}, g_{seg})$). When segmentation cross-entropy generates steep initial gradients, dynamic balancing schemes like GradNorm can destabilize shared encoder updates; in such regimes, static loss weighting ratios between 5:1 and 10:1 provide substantially superior convergence stability.

VII. Conclusion and Future Outlook

This paper presented a systematic empirical inquiry into hard parameter-sharing multi-task architectures across four clinical imaging cohorts. We characterized foundational evaluation, resolution, and optimization boundaries in medical multi-task vision: we proved the empty-mask Dice baseline floor analytically and empirically, demonstrated that spatial downsampling separates proxy patch categorization from clinical ordinal grading in digital pathology, and identified gradient instability under dynamic balancing. Furthermore, we showed that standard color deconvolution can attenuate fine cellular chromatin features, while lateral skip connections exert minimal influence on shared classification representations. Together with our five benchmarking guidelines, these findings establish robust methodological foundations to ensure clinically trustworthy multi-task medical vision systems.

Future work will investigate integrating multi-scale attention mechanisms, exploring adaptive gradient projection operators (e.g., PCGrad [22]), and leveraging frozen foundation model representations (e.g., UNI [3], CONCH [4]) to bridge the resolution gap between computational constraints and high-resolution clinical pathology. To support scientific replication and community standardization, all code, patient-isolated cross-validation fold splits, and evaluation pipelines are openly available at https://github.com/ApatheticMioz/cancer-pathology-dl.

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